Dynamic Resolution for Task-Adaptive Efficient 3D Navigation and Target Detection

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CS-503 Project Proposal

***Abstract*—Current 3D navigation agents use fixed-resolution images, forcing a trade-off between computational cost and perceptual detail: low resolution limits object recognition, while high resolution is expensive to process at every step. We propose a reinforcement learning (RL) agent in AI2-THOR that can increase its observation resolution via a sensing action under a limited perceptual budget, enabling adaptive vision that is efficient for both navigation and target recognition.**

I. Introduction

Object search in 3D scenes is a common task in embodied AI. An agent must move through an environment, interpret images, and decide where to go in order to find an object.

In this project, we consider a setting where perception itself is an active and constrained process. The agent initially observes the scene at low resolution and may choose to acquire additional information by increasing the resolution of its current view before acting.

This project is motivated by two main observations. First, biological vision operates under similar constraints: perception is limited in resolution and attention is required to focus on relevant parts of the scene, which can be implicitly modeled through dynamic resolution. Second, selectively acquiring information can be more efficient than processing high-resolution images at every step, and allows the agent to adapt its perception to different stages of the task, such as navigation versus object recognition.

A. What is the problem you want to solve?

Most 3D navigation agents are trained on fixed resolution images, but it might not be suited for all aspects of the task.

Prior work [1] has shown that agents can perform simple navigation tasks even with very low resolution observations. However, these agents could struggle in more complex environments, e.g. when contrast is low or when objects are numerous/complex. Navigation can be achieved with low visual information, but object recognition should require more details to be precise. Moreover, continuously processing high resolution images is computationally heavy and unnecessary in many situations. So there is a gap to bridge to have a model effective in both tasks and low cost.

We therefore propose to explore a similar setting by introducing dynamic resolution, allowing the agent to improve its camera resolution over time when it chooses to focus on the scene currently in vision. This is also a more natural approach to vision since it implicitly reflects attention. In real settings with robots, this can be implemented with multi-frame super-resolution, where multiple subpixel-shifted low-resolution images are combined to reconstruct a higher-resolution observation [2].

B. Why is it important that this problem be solved?

Many real world systems operate under constraints on sensing and computation. A realistic agent should not rely on constant access to very dense information, but instead learn when it should acquire additional information.

- **More realistic perception modeling.** The progressive observation model captures the fact that perception is limited in resolution and constrained by time/energy, making the agent's vision closer to biological and real-world sensing.
- **Computational efficiency.** By learning when high-resolution observations are truly necessary, the agent can reduce the number of expensive sensing and processing operations.
- **Robustness in complex environments.** We believe that low resolution models, even if they demonstrated strong capacity, might struggle in more complex environments, and dynamic resolution could bridge this gap.

II. Method and Deliveries

A. How do you solve the problem?

We consider a standard RL setup for 3D navigation using the AI2-THOR simulator [3]. Each episode begins with a target object and an initial agent pose. The goal is to navigate the environment, reach a target object, and center it in the field of view. The environment consist of apartment scenes.

The agent can perform standard navigation actions (*move forward*, *rotate left*, *rotate right*, *stop*), as well as a dedicated sensing action. This sensing action allows the agent to increase the resolution of its current observation while remaining at the same physical location. Once the agent moves, the observation resets to low resolution.

The episode ends when the agent is either successful in the task or has spent too much time in the simulation. We will also define a limited perceptual "budget" and a perceptual cost, to really force the model to only increase resolution when necessary, but also to have a fair limitation between different agent behavior (see below).

The agent observations consist of RGB inputs processed by a visual encoder and the list of previous action taken with addition of a GPS+Compass sensor as it is also standard and done in [1]. The visual encoder will be a LSTM since it seems to be standard for navigation task with partially observable information [4], and also because there is no point in using dynamic resolution if the model does not possess some kind of memory. The policy outputs either navigation actions or resolution augmentation.

The reward function will encourage successful target finding while penalizing unnecessary time steps. A simple design that we could use is: positive terminal reward for success, positive reward depending on distance to target object, negative reward for timeout or failure to remain in budget, small step penalty, an optional perceptual cost penalty to further discourage useless observations, and a medium positive reward for exploring new rooms (still in discussion for the exact implementation).

Figure 1. Illustration of AI2-THOR scene diversity and complexity

B. How will you validate your solution?

We will compare our adaptive agent against baselines:

- **Low-resolution agent:** no sensing action available.
- **High-resolution agent:** always observes the environment at maximum resolution.
- **Fixed sensing schedule:** one high-res sensing occurs at predefined intervals.
- **Random sensing:** sensing actions are randomly selected under the same budget.

Performance will be evaluated using the below metrics:

- **Classic Task performance:** success rate, navigation efficiency, episode length, Success weighted by Path Length (SPL), distance-to-goal reduction, convergence speed and reward evolution.
- **Perception efficiency:** number of sensing actions, summed sensing cost, performance under fixed budgets.
- **Behavioral analysis:** when and where sensing is used, how frequently, and perhaps interpretation.

C. Timeline

The most time consuming part will likely be fine tuning the reward function, as it is difficult to predict how the model will adapt to dynamic resolution. We want to penalize excessive sensing, but not to the point where the agent avoids using it entirely or uses it too conservatively. This will likely require several iterations.

A realistic timeline would be to set up the full learning environment by week 3 (e.g., implementing the LSTM and integrating it with the AI2-THOR controller). This would be followed by one to two weeks of reward tuning, then training the different models, and finally at least one week dedicated to generating visualizations and analyzing results.

Additional experiments could be conducted during or after training if time allows. For example, we can vary the sensing budget (low, medium, high), and evaluate the robustness of the model by modifying the environment settings, such as room size (small vs medium), object density (few vs many objects), or lighting conditions (darker vs brighter scenes).

III. Related Work

As cited before, [1] already experiments with low resolution agent, which is our main inspiration and where we get most of our insight on the model. [5] explores in a similar way what we want to demonstrate, where a cheap selector model is trained to select the lowest resolution for a subsequent CNN model to correctly perform image classification. This highlights that full resolution images are not always necessary in vision models.

To our knowledge, there is no directly comparable setup that introduces dynamic resolution in a RL agent.

IV. Discussion

A. Implications and Contribution

We hope to demonstrate that dynamic resolution models can be as effective as high resolution models, while being more cost efficient. This could show that navigation models and robots with limited resources can be optimized through super-resolution. It could also provide an other approach to modeling attention in vision-based learning systems.

B. Feasibility Assessment

- **Data:** All scenes are available through the AI2-THOR framework, and additional ones can be generated if needed, which is one reason for choosing this simulator.
- **Software tools:** AI2-THOR provides the environment and an agent controller with a convenient API, which we will extend to implement the RL model with dynamic vision.
- **Izar:** AI2-THOR appears to run on the Izar clusters, but the required computational budget is still uncertain due to unknown convergence time and simulation cost.
- **Risks:** AI2-THOR may be resource-intensive due to physics and scene complexity, leading to long training times. This can be mitigated by disabling some physics and limiting object types. Another risk is that reward tuning may take longer than expected.

References

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